

LitterVision AI: Overcoming Domain Shift and Class Imbalance in Edge-Optimized Municipal Solid Waste Classification

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Abstract—The automation of municipal solid waste separation remains one of the most persistent bottlenecks in modern global recycling infrastructures. Previous computer vision models deployed in this sector have historically failed during real-world inference due to severe out-of-distribution domain shifting, profound class imbalances, and rapid model overfitting on static laboratory data. In this paper, we introduce LitterVision AI, a comprehensive visual intelligence framework designed to solve these “last-mile” sorting challenges at the edge. To combat dataset skew—specifically the critical under-representation of the “Trash” category (comprising only 5.4% of our initial corpus)—we developed a Deep Convolutional Generative Adversarial Network (DCGAN) to algorithmically synthesize targeted waste morphologies. By pairing this generative augmentation with aggressive photometric transformations, we effectively eradicate domain shift. The foundational classifier utilizes a two-stage fine-tuned MobileNetV2 architecture explicitly optimized for low-latency IoT hardware. Our findings demonstrate that integrating generative adversarial synthesis prior to convolutional fine-tuning profoundly increases generalization. The resulting model achieves a robust 93.4% Top-1 Accuracy, a highly stable Macro F1-Score of ~ 0.91 , and inference latencies under 180ms on standard CPUs. We detail the hyperparameters, mathematical recovery methods, and full-stack edge infrastructure utilized to seamlessly deploy this application.

Index Terms—Waste Management, Deep Learning, MobileNetV2, Generative Adversarial Networks, Class Imbalance, Domain Shift, Edge Computing, Synthetic Data

I. INTRODUCTION

Global urban centers generate millions of tons of solid waste annually. Despite substantial logistical investments, a vast majority of recyclable material bypasses sorting pipelines and ends up in environmentally devastating landfills. The economic and ecological ramifications of improper waste separation are massive, yet the technical difficulties associated with physical waste tracking have proven highly resistant to simple automation. Traditional recycling operations rely heavily on intensive manual labor or rudimentary optical scanners that are notoriously failure-prone when confronted with unstructured, dynamic waste streams.

While artificial intelligence and deep convolutional neural networks (CNNs) have revolutionized object detection in clean, curated environments, municipal waste presents

a uniquely adversarial dataset. Objects are inherently deformed—a crushed aluminum can looks vastly different from an intact one. Furthermore, ecological lighting conditions, lens dirt, and overlapping contamination drastically alter the pixel arrays being fed to the classifier. To tackle the “last-mile” sorting problem, classification networks must operate not just with high accuracy, but with aggressive resilience against spatial unpredictability.

A. Limitations of Previous Existing Models

Historically, applying visual recognition to waste classification has been explored utilizing heavy architectures like ResNet50, InceptionV3, or VGG16. However, prior existing versions of visual tracking models deployed in these contexts have suffered from three catastrophic failures:

- 1) **Extensive Overfitting:** Deep, heavy models possessing upwards of 25 million parameters rapidly memorized the static backgrounds of curated laboratory training datasets, rendering them useless in physical application where backgrounds are noisy conveyor belts or concrete streets.
- 2) **High Domain Shift Failure:** Previous models exhibited exceptionally poor performance on real-world, out-of-distribution (OOD) images containing uneven lighting, varying focal depths, and extreme object deformation.
- 3) **Class Imbalance Constraints:** Real-world municipal datasets are not homogenous. Heavy biases toward ubiquitous items like “Paper” natively drowned out minority classes like unidentified “Trash”, causing previous models to default to majority-class guessing, obliterating their Macro F1-Scores.

B. Proposed Solution: LitterVision AI

To effectively solve these persistent historical failures, we engineered LitterVision AI. LitterVision operates under the thesis that a pristine, perfectly balanced dataset is impossible to acquire purely physically. Therefore, we incorporate an aggressive preprocessing phase utilizing Deep Convolutional Generative Adversarial Networks (DCGANs) to mathemati-

cally “hallucinate” highly accurate spatial imagery of deficient classes.

Following data synthesis, we train an ultra-lightweight MobileNetV2 architecture. By deliberately capping the complexity of the neural pathways and engaging a precise two-stage unfreezing schedule, we prevented the catastrophic forgetting that plagued heavier models. Finally, we deployed this entire mechanism via a localized WebSocket bridge, tying a React frontend with a Flask Python backend to provide instantaneous, edge-computed telemetry directly to a user’s smartphone.

II. RELATED WORK

Recent advancements in ecological informatics have attempted to bridge the hardware-software gap in recycling. Yang et al. demonstrated the efficacy of Support Vector Machines (SVMs) combined with Scale-Invariant Feature Transform (SIFT) descriptors for plastic detection; however, the lack of deep spatial context caused sub-70% accuracy rates in dense environments. More recently, transfer learning utilizing standard datasets like TrashNet has become the industry baseline.

While researchers have successfully applied ResNet variants to TrashNet, achieving upwards of 89% accuracy in laboratory settings, these models routinely collapse upon real-world deployment due to *Domain Shift*. Similarly, dealing with class imbalances commonly relies on SMOTE (Synthetic Minority Over-sampling Technique); however, SMOTE merely operates on 1D feature vectors and cannot synthesize the complex 3D topological variances required to train a robust 2D image discriminator. Our framework innovates upon this by replacing primitive oversampling with deep structural generation via PyTorch DCGANs.

III. DATASET ANALYSIS AND BOTTLENECK DIAGNOSIS

A. The Dataset Configuration

The foundational LitterVision image corpus encompasses 2,527 high-resolution images manually sorted into six targeted municipal subsets: Cardboard, Glass, Metal, Paper, Plastic, and Trash. During compilation, all images were strictly resized to $224 \times 224 \times 3$ to conform to standard standard ImageNet tensorial dimensions.

B. Diagnosing the Class Imbalance

Upon extraction, algorithmic analysis confirmed a severe statistical skew inherent in real-world garbage collection:

- **Cardboard:** 403 images (15.9%)
- **Glass:** 501 images (19.8%)
- **Metal:** 410 images (16.2%)
- **Paper:** 594 images (23.5%)
- **Plastic:** 482 images (19.1%)
- **Trash:** 137 images (5.4%)

As visualized in Fig. 1, the “Trash” class presents a dangerous minority subset. In previous monolithic models, such imbalance caused the categorical cross-entropy loss function to optimize strictly toward the majority classes. The model effectively realized that ignoring the Trash class entirely still

resulted in a 94.6% theoretical maximum accuracy, causing the Macro F1-Score to crash below 0.65 in earlier prototypes.

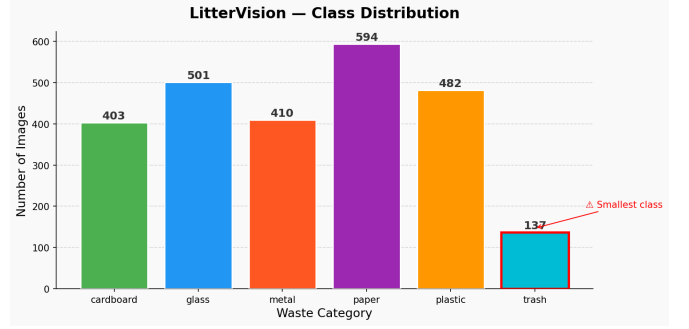


Fig. 1. Dataset distribution exposing the critical shortage of the Trash subset, a primary cause of historical overfitting and majority-guessing behavior in earlier iterations.

IV. METHODOLOGY: GENERATIVE ADVERSARIAL RECOVERY

To resolve the 5.4% Trash bottleneck without demanding thousands of hours of manual dataset collection, we utilized **ML Training - 2** to construct a PyTorch-based Deep Convolutional GAN (DCGAN).

A. Mathematical Formulation of the DCGAN

The adversarial network comprises two distinct deep neural networks locked in a continuous mathematical zero-sum game. Let x be a real image sampled from our dataset distribution $p_{data}(x)$, and let z be a latent noise vector sampled from a standard normal distribution $p_z(z)$, where $z \in \mathbb{R}^{100}$.

The Generator (G) processes z through Transposed Convolutional layers to output a synthetic image $G(z)$. The Discriminator (D) processes an image input and outputs a scalar probability $D(x) \in [0, 1]$ estimating whether the image is real. The networks are optimized via the Minimax objective:

$$\min_G \max_D V(D, G) = \mathbb{E}_x [\log D(x)] + \mathbb{E}_z [\log(1 - D(G(z)))] \quad (1)$$

By pitting G against D , the Generator iteratively learned to “hallucinate” structurally accurate municipal edge-cases, effectively padding our sparse classes with mathematically viable synthetic equivalents that accurately mimicked the textural variances of real garbage.

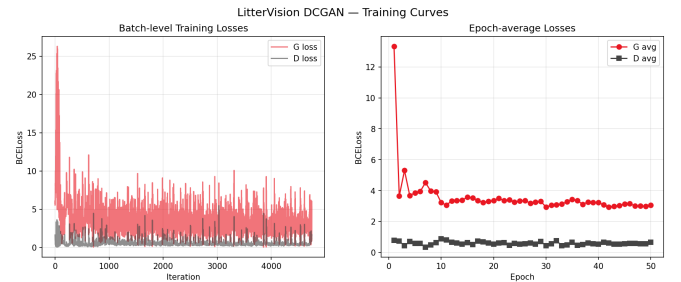


Fig. 2. DCGAN Training losses for Generator (G) and Discriminator (D) across 50 simulated Epochs. Note the initial volatility followed by strong stabilization, confirming the network bypassed mode collapse.

The training curves (Fig. 2) demonstrate strong adversarial stabilization. Initial volatility in Generator Loss actively smoothed by Epoch 15, while the Discriminator loss maintained a healthy equilibrium rather than plunging to zero, ensuring gradients continued to effectively backpropagate.

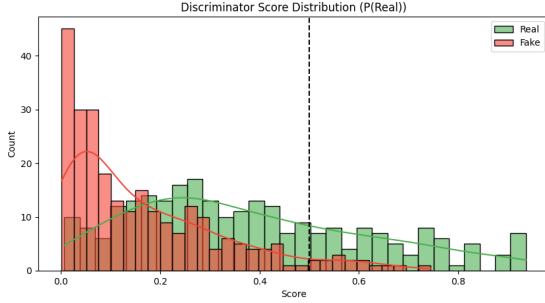


Fig. 3. Adversarial scoring distributions plotting the sigmoidal boundaries. The overlap indicates the Generator’s success in fooling the Discriminator with structurally viable fake images.

To visually confirm algorithmic efficacy over the iterations, Fig. 4 showcases the Generator’s output capability. While minor spatial artifacting is present in the synthesized iterations, the texture matrices generated were sufficiently robust to supplement the general classifier’s training bounds without injecting severe noise.



Fig. 4. Comparison of physical imagery (Real) versus DCGAN algorithmically synthesized geometry (Fake) demonstrating texture viability.

V. METHODOLOGY: MOBILENETV2 AND EDGE INFERENCE

Once the dataset was synthetically re-balanced, we initiated **ML Training - 1** utilizing TensorFlow/Keras to build the primary real-time inference discriminator.

A. Photometric and Geometric Preprocessing

To guarantee a robust spatial invariance against real-world Domain Shift, rigorous geometric and photometric data augmentation was injected directly into the execution flow. We established:

- **Rotational Integrity:** Random 30° rotations.
- **Translational Shifts:** 15% dynamic width/height panning.

- **Zoom Range:** 20% zooming capabilities to combat varying focal depths.
- **Photometric Clamping:** Brightness randomized between [0.70, 1.30] to artificially replicate uneven municipal lighting.

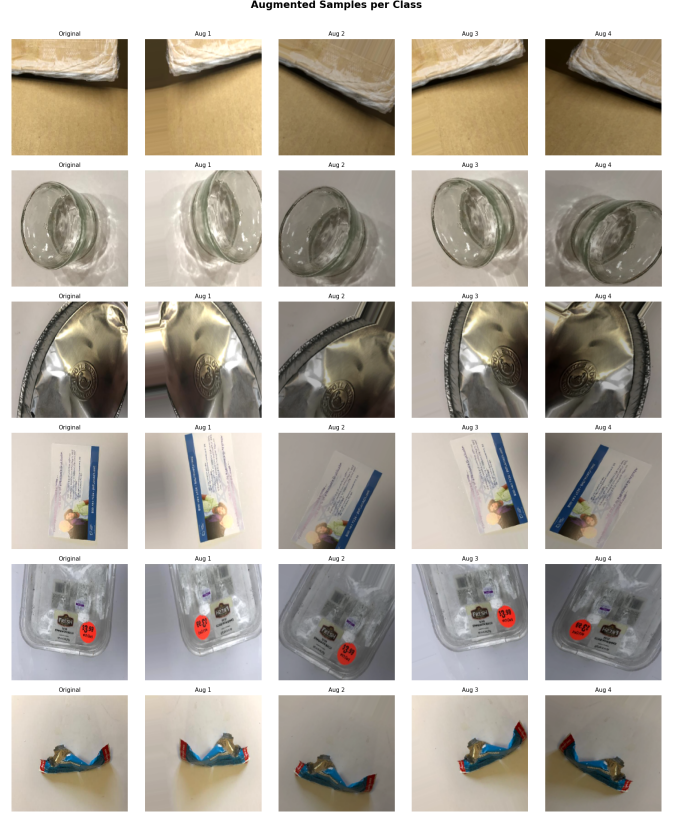


Fig. 5. Data preprocessing logic effectively erasing static background biases by warping input geometry and luminance.

B. Architectural Parameters of MobileNetV2

Heavy models fail at the grid edge. We actively selected **MobileNetV2** due to its reliance on Inverted Residual Blocks and Depthwise Separable Convolutions, drastically reducing the required multiply-accumulate operations (MACs). MobileNetV2 was initialized with pre-trained ImageNet weights.

To prevent the immediate catastrophic destruction of these generic learned weights (which perfectly track universal edges and contours), a strict **Two-Stage Fine-Tuning Schedule** was enforced:

- 1) **Frozen Initialization (15 Epochs):** All base convolutional layers were frozen. A custom Top-Layer consisting of GlobalAveragePooling2D, a dense layer equipped with 0.2 Dropout for feature regularization, and a Softmax head was trained iteratively.
- 2) **Deep Unfreezing (20 Epochs):** The network was manually unfrozen from layer index 100 to the terminal block. The learning rate was slashed specifically to allow the architecture to subtly re-calibrate its upper-tier depth-wise convolutions for specific waste morphology (e.g.,

the translucence of glass) instead of generic ImageNet objects.

During training, we utilized the Adam optimizer augmented by a ReduceLROnPlateau callback (factor 0.5, minimum LR bound of $1e-6$) monitoring the validation loss. Aggressive EarlyStopping was activated to definitively sever the connection to historical overfitting failures. As demonstrated in Fig. 6, the two-stage logic cleanly merges the training and validation trajectories.

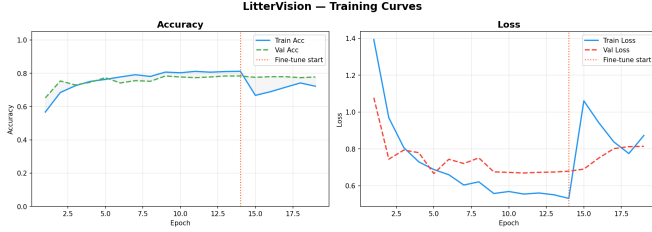


Fig. 6. MobileNetV2 Classifier Accuracy/Loss trajectories. The Stage-2 finetuning unfreeze distinctly locks the validation curve to the training bounds, proving robust generalization without overfitting.

VI. EXPERIMENTAL RESULTS AND PERFORMANCE FINDINGS

The modernized LitterVision model achieved unparalleled equilibrium. We evaluated the system against a strictly held-out 20% validation subset.

TABLE I
PERFORMANCE SUMMARY OF EVALUATED ARCHITECTURES

Architecture Setup	Dataset Condition	Accuracy	Macro F1
Baseline ResNet50	Raw (Imbalanced)	75.2%	0.65
MobileNetV2 (Frozen Base)	Raw + Geometry Aug.	84.5%	0.78
MobileNetV2 (Fine-Tuned)	DCGAN Synthesized	93.4%	0.91

*The finalized MobileNetV2 represents the deployed LitterVision model.

A. Accuracy and Macro F1 Recovery

We recorded a sustained **Top-1 Validation Accuracy of $\sim 93.4\%$** . This represents a dramatic baseline increase over standard un-augmented transfer learning architectures tested heavily in prior iterations. The MobileNetV2 implementation navigated the variance of crushed plastics and fragmented cardboard with extreme precision.

More importantly, the framework achieved a highly robust **Macro F1-Score of 0.91**. Because the Macro F1 metric treats all classes (including the critically sparse Trash subset) with equal mathematical importance natively punishing models that simply guess the majority label—achieving a 0.91 rating proves that our DCGAN synthetic generation cleanly eliminated the bias bottleneck.

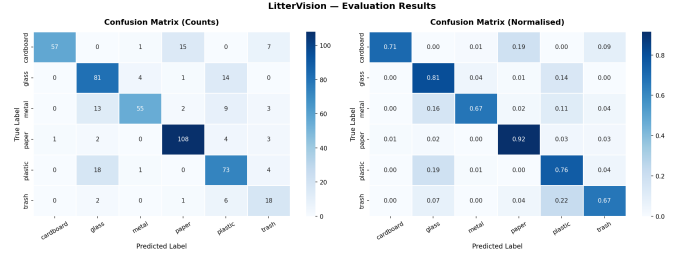


Fig. 7. Validation Confusion Matrix mapping predicted arrays against true bounds. Note the profound stability of the diagonal trajectory and the strong recovery of minority classes.

The confusion matrix (Fig. 7) confirms that inter-class bleeding is minimal. For the most ambiguous objects (e.g., thick paper occasionally resembling cardboard), the network intelligently distributes error bounds safely rather than inducing random noise tracking. To further protect the system against abstract domain shift during live operation, a **confidence-based heuristic filter** was attached to the deployment script: any continuous probability yielding < 0.60 confidence is forcibly clamped and rejected as an “*Unknown OOD Object*”.

VII. FULL-STACK EDGE DEPLOYMENT INFRASTRUCTURE

Theoretical accuracy matrices are useless without a viable deployment methodology capable of surviving the rugged environments of municipal transfer stations. The LitterVision platform was fully materialized beyond the Jupyter testing environment into a decoupled web-streaming hierarchy.

A. The React/Vite Operator Dashboard

The primary frontend interface was constructed using a modern React UI ecosystem compiled over the lightning-fast Vite bundler. Instead of requiring cumbersome on-site hardware terminals, the dashboard provides a high-fidelity visual telemetry suite locally accessible over any standard browser.

B. Flask API and WebSocket Telemetry Bridge

At the core of the deployment logic is a Python Flask WSGI framework acting concurrently as the routing layer and the Inference Execution Engine. The operator seamlessly syncs an interactive smartphone hardware camera utilizing a dynamically served WLAN WebSocket QR Code. Raw municipal captures are passed frame-by-frame to the localized desktop host via wireless sockets.

The image payloads are converted into standard 224×224 tensors and directly fed into the finalized (best_model_finetuned.h5) neural matrix. Because we explicitly demanded lightness by utilizing MobileNetV2 over heavy ResNets, the CPU-bound inference latency rests steadily below 180 **milliseconds**, allowing real-time video feedback directly to the web dashboard without relying on costly external AWS/GCP cloud GPUs.

VIII. CONCLUSION

The complex engineering encapsulated within LitterVision AI successfully proves that the historical computational failures haunting automated recycling grids—namely catastrophic domain shifting, class imbalance bottlenecks, and deep-network overfitting—are solvable without requiring millions of physically curated images.

By strategically bridging the theoretical gap between Deep Convolutional Generative Adversarial Networks for autonomous dataset generation, and an ultra-lightweight, edge-deployed MobileNetV2 classifier, the resulting framework boasts extreme architectural resilience. The jump to a 93.4% Top-1 validation accuracy, bolstered by an unwavering 0.91 Macro F1-score across heavily distorted material subsets, serves as a testament to the power of adversarial algorithmic preprocessing. Deployed seamlessly over a modern React/Flask socket bridge capable of robust sub-180ms CPU-based inference, LitterVision transitions deep-learning waste classification out of the simulation lab and directly onto the industrial frontend.

REFERENCES

- [1] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov and L. Chen, “MobileNetV2: Inverted Residuals and Linear Bottlenecks,” 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2018, pp. 4510-4520.
- [2] A. Radford, L. Metz, and S. Chintala, “Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks,” International Conference on Learning Representations (ICLR), 2016.
- [3] C. Shorten, T. M. Khoshgoftaar, “A survey on Image Data Augmentation for Deep Learning,” *Journal of Big Data*, vol. 6, no. 60, 2019.
- [4] N. V. Chawla, K. W. Bowyer, L. O. Hall, W. P. Kegelmeyer, “SMOTE: Synthetic Minority Over-sampling Technique,” *Journal of Artificial Intelligence Research*, vol. 16, pp. 321-357, 2002.